

HDOC Day 24 Report Programming in C

PROBLEM

Write a C program that displays the following menu and uses a switch statement to print the “Increasing patterns”

selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate

parameterless function with no return value.

Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program,

and test it against the prepared test cases.

1. Increasing right-angled triangle

Input the number of rows, n, and print an increasing right-angled triangle using asterisks.

For n = 4:

*

* *

* * *

* * * *

2. Increasing row-number triangle

Input the number of rows, n, and print a triangle in which each row number is printed as many times as its

value.

For n = 4:

1

2 2

3 3 3

4 4 4 4

3. Increasing natural-number pattern

Input the number of rows, n, and print natural numbers from 1 to the corresponding row number in each row.

For n = 4:

1

1 2

1 2 3

1 2 3 4

4. Consecutive-number triangle

Input the number of rows, n, and print consecutive natural numbers in a triangular pattern.

For n = 4:

1

2 3

4 5 6

7 8 9 10

5. Increasing alphabet pattern

Input the number of rows, n, and print uppercase English letters from A up to the letter corresponding to each

row. Assume that $1 \leq n \leq 26$ for the alphabet pattern.

For n = 4:

A

A B

A B C

A B C

ALGORITHM

MAIN Function

Step 1: Start

Step 2: Display menu options (1 to 5)

Step 3: Read user input into 'choice'

Step 4: Evaluate 'choice' using switch statement:

Case 1: Execute pattern1()

Case 2: Execute pattern2()

Case 3: Execute pattern3()

Case 4: Execute pattern4()

Case 5: Execute pattern5()

Default: Print "Invalid choice"

Step 5: End

PATTERN1 function

Step 1: Start

Step 2: Read number of rows 'n'

Step 3: Set outer loop i = 1

Step 4: If i > n, go to Step 9

Step 5: Set inner loop j = 1

Step 6: If j > i, print newline (\n), increment i by 1, and go to Step 4

Step 7: Print "*" "

Step 8: Increment j by 1 and go to Step 6

Step 9: End

PATTERN 2 FUNCTION

Step 1: Start

Step 2: Read number of rows 'n'

Step 3: Set outer loop i = 1

Step 4: If i > n, go to Step 9

Step 5: Set inner loop j = 1

Step 6: If j > i, print newline (\n), increment i by 1, and go to Step 4

Step 7: Print value of 'i' followed by a space

Step 8: Increment j by 1 and go to Step 6

Step 9: End

PATTERN3 FUNCTION

Step 1: Start

Step 2: Read number of rows 'n'

Step 3: Set outer loop $i = 1$

Step 4: If $i > n$, go to Step 9

Step 5: Set inner loop $j = 1$

Step 6: If $j > i$, print newline ($\backslash n$), increment i by 1, and go to Step 4

Step 7: Print value of 'j' followed by a space

Step 8: Increment j by 1 and go to Step 6

Step 9: End

PATTERN4 FUNCTION

Step 1: Start

Step 2: Read number of rows 'n'

Step 3: Initialize counter variable $count = 1$

Step 4: Set outer loop $i = 1$

Step 5: If $i > n$, go to Step 11

Step 6: Set inner loop $j = 1$

Step 7: If $j > i$, print newline ($\backslash n$), increment i by 1, and go to Step 5

Step 8: Print value of 'count' followed by a space

Step 9: Increment $count$ by 1

Step 10: Increment j by 1 and go to Step 7

Step 11: End

C PROGRAM CODE

```
#include <stdio.h>
```

```
void pattern1(void);
```

```
void pattern2(void);
```

```
void pattern3(void);
```

```
void pattern4(void);
```

```
void pattern5(void);
```

```
int main(void) {
```

```
    int choice;
```

```
    printf("=====\n");
```

```
    printf("    INCREASING PATTERNS MENU    \n");
```

```
    printf("=====\n");
```

```
    printf("1. Increasing right-angled triangle\n");
```

```
    printf("2. Increasing row-number triangle\n");
```

```
    printf("3. Increasing natural-number pattern\n");
```

```
    printf("4. Consecutive-number triangle\n");
```

```
    printf("5. Increasing alphabet pattern\n");
```

```
    printf("=====\n");
```

```
    printf("Enter your choice (1-5): ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice) {
```

```
        case 1:
```

```
            pattern1();
```

```
            break;
```

```
        case 2:
```

```
            pattern2();
```

```
            break;
```

```
        case 3:
```

```
            pattern3();
```

```
        break;
    case 4:
        pattern4();
        break;
    case 5:
        pattern5();
        break;
    default:
        printf("\nInvalid choice! Please select an option between 1 and 5.\n");
}

return 0;
}
```

```
void pattern1(void) {
    int n, i, j;
    printf("\nEnter number of rows (n): ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }
}
```

```
void pattern2(void) {
```

```
int n, i, j;

printf("\nEnter number of rows (n): ");

scanf("%d", &n);

for (i = 1; i <= n; i++) {
    for (j = 1; j <= i; j++) {
        printf("%d ", i);
    }
    printf("\n");
}
```

```
void pattern3(void) {
    int n, i, j;

    printf("\nEnter number of rows (n): ");

    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
}
```

```
void pattern4(void) {
    int n, i, j, count = 1;

    printf("\nEnter number of rows (n): ");
```

```
scanf("%d", &n);
```

```
for (i = 1; i <= n; i++) {  
    for (j = 1; j <= i; j++) {  
        printf("%d ", count);  
        count++;  
    }  
    printf("\n");  
}  
}
```

```
void pattern5(void) {  
    int n, i, j;  
    printf("\nEnter number of rows (n, 1 to 26): ");  
    scanf("%d", &n);  
  
    if (n < 1 || n > 26) {  
        printf("Invalid input! n must be between 1 and 26.\n");  
        return;  
    }  
  
    for (i = 1; i <= n; i++) {  
        for (j = 0; j < i; j++) {  
            printf("%c ", 'A' + j);  
        }  
        printf("\n");  
    }  
}
```